

Federated Breast Cancer Subtype Classification

Final results summary

generated 2026-08-05 10:04 UTC

[done]	test01	test01_centralized	macro-AUC	0.6068
[done]	test02	test02_fedavg_2h	macro-AUC	0.5594
[done]	test03	test03_fedprox_2h	macro-AUC	0.5917
[done]	test04	test04_fedavg_3h	macro-AUC	0.5990
[done]	test05	test05_fedprox_3h	macro-AUC	0.5958
[done]	test06	test06_fedavg_4h	macro-AUC	0.6531
[done]	test07	test07_fedprox_4h	macro-AUC	0.6075
[done]	test08	test08_fedavg_skewed	macro-AUC	0.5982
[done]	test09	test09_fedprox_skewed	macro-AUC	0.6250
[done]	test10	test10_fedavg_cohort	macro-AUC	0.5426
[done]	test12	test12_fedavg_sizematched	macro-AUC	0.5836
[done]	test11	test11_fedprox_cohort	macro-AUC	0.5678
[done]	test13	test13_fedprox_sizematched	macro-AUC	0.5882

Main Results (columns 1/4)

experiment	run	status	datetime	model	dataset	hospitals		partition	split
test01	test01_centralized	complete	2026-08-03T23:48:43+00:00	resnet18	multi_subtype_80mm	1		-	all data pooled
test02	test02_fedavg_2h	complete	2026-08-04T00:17:29+00:00	resnet18	multi_subtype_80mm	2		2_clients_balanced	2 hospitals, balanced (50/50) -> 50.0%
test03	test03_fedprox_2h	complete	2026-08-04T00:23:45+00:00	resnet18	multi_subtype_80mm	2		2_clients_balanced	2 hospitals, balanced (50/50) -> 50.0%
test04	test04_fedavg_3h	complete	2026-08-04T00:29:22+00:00	resnet18	multi_subtype_80mm	3		3_clients_balanced	3 hospitals, balanced (33.3 each) -> 33.3% / 33.3%
test05	test05_fedprox_3h	complete	2026-08-04T00:35:20+00:00	resnet18	multi_subtype_80mm	3		3_clients_balanced	3 hospitals, balanced (33.3 each) -> 33.3% / 33.3%
test06	test06_fedavg_4h	complete	2026-08-04T00:41:01+00:00	resnet18	multi_subtype_80mm	4		4_clients_balanced	4 hospitals, balanced (25 each) -> 25.0% / 25.0%
test07	test07_fedprox_4h	complete	2026-08-04T00:47:01+00:00	resnet18	multi_subtype_80mm	4		4_clients_balanced	4 hospitals, balanced (25 each) -> 25.0% / 25.0%
test08	test08_fedavg_skewed	complete	2026-08-04T00:52:49+00:00	resnet18	multi_subtype_80mm	4	4 hospitals skewed	4_clients_skewed	4 hospitals, skewed (1 dissertation: 50/20/10/10) -> 50.0%
test09	test09_fedprox_skewed	complete	2026-08-04T00:59:41+00:00	resnet18	multi_subtype_80mm	4	4 hospitals skewed	4_clients_skewed	4 hospitals, skewed (1 dissertation: 50/20/10/10) -> 50.0%
test10	test10_fedavg_cohort	complete	2026-08-05T01:16:51+00:00	resnet18	multi_subtype_80mm	3		3_hospitals_cohort	3 hospitals, same cohort each (DUKE I-SPY1 I-SPY2) -> 33.3%
test12	test12_fedavg_sizematched	complete	2026-08-05T01:25:10+00:00	resnet18	multi_subtype_80mm	3		3_hospitals_sizematched	3 hospitals, sizes matched to 3_clients_cohort
test11	test11_fedprox_cohort	complete	2026-08-05T01:33:40+00:00	resnet18	multi_subtype_80mm	3		3_hospitals_cohort	3 hospitals, same cohort each (DUKE I-SPY1 I-SPY2) -> 33.3%
test13	test13_fedprox_sizematched	complete	2026-08-05T01:42:13+00:00	resnet18	multi_subtype_80mm	3		3_hospitals_sizematched	3 hospitals, sizes matched to 3_clients_cohort

Main Results (columns 2/4)

algorithm	seed	best epoch	best round	train time (s)	time measured as	model used	test patients	trivial baseline
centralized	42	4		268	sum of per-epoch compute	selected	268	0.5112
fedavg	42		25	313	job wall clock (incl. orchestration)	selected	268	0.5112
fedprox	42		29	351	job wall clock (incl. orchestration)	selected	268	0.5112
fedavg	42		27	313	job wall clock (incl. orchestration)	selected	268	0.5112
fedprox	42		28	333	job wall clock (incl. orchestration)	selected	268	0.5112
fedavg	42		0	317	job wall clock (incl. orchestration)	selected	268	0.5112
fedprox	42		0	335	job wall clock (incl. orchestration)	selected	268	0.5112
fedavg	42		21	323	job wall clock (incl. orchestration)	selected	268	0.5112
fedprox	42		2	387	job wall clock (incl. orchestration)	selected	268	0.5112
fedavg	42		7	469	job wall clock (incl. orchestration)	selected	268	0.5112
fedavg	42		17	471	job wall clock (incl. orchestration)	selected	268	0.5112
fedprox	42		16	482	job wall clock (incl. orchestration)	selected	268	0.5112
fedprox	42		27	485	job wall clock (incl. orchestration)	selected	268	0.5112

Main Results (columns 3/4)

accuracy	balanced accuracy	macro precision	macro recall	macro F1	macro AUC	auc HRposHER2neg	auc TripleNeg	auc HER2pos
0.5299	0.4503	0.4606	0.4503	0.4523	0.6068	0.6238	0.6886	0.5079
0.4030	0.3742	0.3915	0.3742	0.3744	0.5594	0.5569	0.6285	0.4928
0.4328	0.4025	0.4028	0.4025	0.4001	0.5917	0.6091	0.6461	0.5199
0.4851	0.4198	0.4333	0.4198	0.4231	0.5990	0.6191	0.6468	0.5311
0.4590	0.4127	0.4208	0.4127	0.4116	0.5958	0.6030	0.6355	0.5488
0.4776	0.4522	0.4646	0.4522	0.4378	0.6531	0.6667	0.6845	0.6082
0.4739	0.4393	0.4389	0.4393	0.4362	0.6075	0.6066	0.6588	0.5572
0.4888	0.4259	0.4374	0.4259	0.4292	0.5982	0.6139	0.6580	0.5227
0.4515	0.4210	0.4365	0.4210	0.4197	0.6250	0.6394	0.6811	0.5545
0.4291	0.3582	0.3523	0.3582	0.3536	0.5426	0.5473	0.6078	0.4728
0.4478	0.4183	0.4187	0.4183	0.4153	0.5836	0.5923	0.6567	0.5019
0.4590	0.4105	0.4302	0.4105	0.4144	0.5678	0.5723	0.6323	0.4988
0.4664	0.3885	0.3925	0.3885	0.3884	0.5882	0.6196	0.6488	0.4963

Main Results (columns 4/4)

f1 HRposHER2neg	f1 TripleNeg	f1 HER2pos	precision HRposHER2neg	precision TripleNeg	precision HER2pos	recall HRposHER2neg	recall TripleNeg	recall HER2pos
0.6486	0.5000	0.2083	0.6038	0.5455	0.2326	0.7007	0.4615	0.1887
0.5179	0.3830	0.2222	0.5702	0.4286	0.1758	0.4745	0.3462	0.3019
0.5197	0.4528	0.2276	0.5641	0.4444	0.2000	0.4818	0.4615	0.2642
0.6197	0.4234	0.2261	0.5986	0.4915	0.2097	0.6423	0.3718	0.2453
0.6038	0.3714	0.2595	0.6250	0.4194	0.2179	0.5839	0.3333	0.3208
0.6015	0.3607	0.3514	0.6202	0.5000	0.2737	0.5839	0.2821	0.4906
0.5833	0.4189	0.3065	0.6063	0.4429	0.2676	0.5620	0.3974	0.3585
0.6127	0.4429	0.2321	0.5918	0.5000	0.2203	0.6350	0.3974	0.2453
0.5594	0.4265	0.2734	0.5887	0.5000	0.2209	0.5328	0.3718	0.3585
0.5563	0.3727	0.1319	0.5374	0.3614	0.1579	0.5766	0.3846	0.1132
0.5447	0.4314	0.2698	0.5833	0.4400	0.2329	0.5109	0.4231	0.3208
0.5704	0.4328	0.2400	0.5643	0.5179	0.2083	0.5766	0.3718	0.2830
0.6075	0.3776	0.1800	0.5705	0.4154	0.1915	0.6496	0.3462	0.1698

Centralized Vs Fedavg (columns 1/2)

experiment	run	algorithm	hospitals	partition	split	accuracy	macro precision	macro recall
test01	test01_centralized	centralized	1	-	all data pooled	0.5299	0.4606	0.4503
test02	test02_fedavg_2h	fedavg	2	2_clients_balanced	2 hospitals, balanced (50/50) -> 50.0%	0.4030	0.3915	0.3742
test04	test04_fedavg_3h	fedavg	3	3_clients_balanced	3 hospitals, balanced (33.3 each) -> 33.3%	0.4851	0.4333	0.4198
test06	test06_fedavg_4h	fedavg	4	4_clients_balanced	4 hospitals, balanced (25 each) -> 25.0% / 25.0%	0.4776	0.4646	0.4522
test08	test08_fedavg_skewed	fedavg	4	4 hospitals skewed 5-2-1-1	4 hospitals skewed 5-2-1-1 (dissertation: 50/20/10/10) -> 50.0% / 20.0% / 10.0% / 10.0%	0.4888	0.4374	0.4259
test10	test10_fedavg_cohort	fedavg	3	3 hospitals, same cohort	3 hospitals, same cohort each (DUKE I-SPY1 I-SPY2)	0.4291	0.3523	0.3582
test12	test12_fedavg_sizematched	fedavg	3	3 hospitals, sizes matched	3 hospitals, sizes matched to 3_clients_cohort	0.4478	0.4187	0.4183

Centralized Vs Fedavg (columns 2/2)

macro F1	macro AUC	best epoch	best round	train time (s)	reference	reference macro AUC	Δ macro AUC	within noise floor
0.4523	0.6068	4		268	centralized (test01)	0.6068	0.0000	yes
0.3744	0.5594		25	313	centralized (test01)	0.6068	-0.0474	yes
0.4231	0.5990		27	313	centralized (test01)	0.6068	-0.0078	yes
0.4378	0.6531		0	317	centralized (test01)	0.6068	0.0463	yes
0.4292	0.5982		21	323	centralized (test01)	0.6068	-0.0086	yes
0.3536	0.5426		7	469	centralized (test01)	0.6068	-0.0642	yes
0.4153	0.5836		17	471	centralized (test01)	0.6068	-0.0232	yes

Centralized Vs Fedprox (columns 1/2)

experiment	run	algorithm	hospitals	partition	split	accuracy	macro precision	macro recall
test01	test01_centralized	centralized	1	-	all data pooled	0.5299	0.4606	0.4503
test03	test03_fedprox_2h	fedprox	2	2_clients_balanced	2 hospitals, balanced (50/50) -> 50.0%	0.4328	0.4028	0.4025
test05	test05_fedprox_3h	fedprox	3	3_clients_balanced	3 hospitals, balanced (33.3 each) -> 33.3%	0.4590	0.4208	0.4127
test07	test07_fedprox_4h	fedprox	4	4_clients_balanced	4 hospitals, balanced (25 each) -> 25.0% / 25.0%	0.4739	0.4389	0.4393
test09	test09_fedprox_skewed	fedprox	4	4 hospitals skewed 5-2-1-1	4 hospitals skewed 5-2-1-1 (dissertation: 50/20/10/10) -> 50.0% / 20.0% / 10.0% / 10.0%	0.4515	0.4365	0.4210
test11	test11_fedprox_cohort	fedprox	3	3 hospitals, same cohort	3 hospitals, same cohort each (DUKE I-SPY1 I-SPY2)	0.4590	0.4302	0.4105
test13	test13_fedprox_sizematched	fedprox	3	3 hospitals, sizes matched	3 hospitals, sizes matched to 3_clients_cohort	0.4664	0.3925	0.3885

Centralized Vs Fedprox (columns 2/2)

macro F1	macro AUC	best epoch	best round	train time (s)	reference	reference macro AUC	Δ macro AUC	within noise floor
0.4523	0.6068	4		268	centralized (test01)	0.6068	0.0000	yes
0.4001	0.5917		29	351	centralized (test01)	0.6068	-0.0151	yes
0.4116	0.5958		28	333	centralized (test01)	0.6068	-0.0110	yes
0.4362	0.6075		0	335	centralized (test01)	0.6068	0.0007	yes
0.4197	0.6250		2	387	centralized (test01)	0.6068	0.0182	yes
0.4144	0.5678		16	482	centralized (test01)	0.6068	-0.0390	yes
0.3884	0.5882		27	485	centralized (test01)	0.6068	-0.0186	yes

Fedavg Vs Fedprox (columns 1/2)

experiment	run	algorithm	hospitals	partition	split	accuracy	macro precision	macro recall
test02	test02_fedavg_2h	fedavg	2	2_clients_balanced	2 hospitals, balanced (50/50) -> 50.0%	0.4030	0.3915	0.3742
test04	test04_fedavg_3h	fedavg	3	3_clients_balanced	3 hospitals, balanced (33.3 each) -> 33.3%	0.4851	0.4333	0.4198
test06	test06_fedavg_4h	fedavg	4	4_clients_balanced	4 hospitals, balanced (25 each) -> 25.0% / 25.0%	0.4776	0.4646	0.4522
test08	test08_fedavg_skewed	fedavg	4	4 hospitals skewed 50/10	4 hospitals skewed 50/10 (dissertation: 50/20/10/10) -> 50.0%	0.4888	0.4374	0.4259
test10	test10_fedavg_cohort	fedavg	3	3 hospitals, same cohort	3 hospitals, same cohort each (DUKE I-SPY1 I-SPY2)	0.4291	0.3523	0.3582
test12	test12_fedavg_sizematched	fedavg	3	3 hospitals, sizes matched	3 hospitals, sizes matched to 3_clients_cohort	0.4478	0.4187	0.4183
test03	test03_fedprox_2h	fedprox	2	2_clients_balanced	2 hospitals, balanced (50/50) -> 50.0%	0.4328	0.4028	0.4025
test05	test05_fedprox_3h	fedprox	3	3_clients_balanced	3 hospitals, balanced (33.3 each) -> 33.3%	0.4590	0.4208	0.4127
test07	test07_fedprox_4h	fedprox	4	4_clients_balanced	4 hospitals, balanced (25 each) -> 25.0% / 25.0%	0.4739	0.4389	0.4393
test09	test09_fedprox_skewed	fedprox	4	4 hospitals skewed 50/10	4 hospitals skewed 50/10 (dissertation: 50/20/10/10) -> 50.0%	0.4515	0.4365	0.4210
test11	test11_fedprox_cohort	fedprox	3	3 hospitals, same cohort	3 hospitals, same cohort each (DUKE I-SPY1 I-SPY2)	0.4590	0.4302	0.4105
test13	test13_fedprox_sizematched	fedprox	3	3 hospitals, sizes matched	3 hospitals, sizes matched to 3_clients_cohort	0.4664	0.3925	0.3885

Fedavg Vs Fedprox (columns 2/2)

macro F1	macro AUC	best epoch	best round	train time (s)	reference	reference macro AUC	Δ macro AUC	within noise floor
0.3744	0.5594		25	313	FedAvg (mean over configurati	0.5893	-0.0299	yes
0.4231	0.5990		27	313	FedAvg (mean over configurati	0.5893	0.0097	yes
0.4378	0.6531		0	317	FedAvg (mean over configurati	0.5893	0.0638	yes
0.4292	0.5982		21	323	FedAvg (mean over configurati	0.5893	0.0089	yes
0.3536	0.5426		7	469	FedAvg (mean over configurati	0.5893	-0.0467	yes
0.4153	0.5836		17	471	FedAvg (mean over configurati	0.5893	-0.0057	yes
0.4001	0.5917		29	351	FedAvg (mean over configurati	0.5893	0.0024	yes
0.4116	0.5958		28	333	FedAvg (mean over configurati	0.5893	0.0065	yes
0.4362	0.6075		0	335	FedAvg (mean over configurati	0.5893	0.0182	yes
0.4197	0.6250		2	387	FedAvg (mean over configurati	0.5893	0.0357	yes
0.4144	0.5678		16	482	FedAvg (mean over configurati	0.5893	-0.0215	yes
0.3884	0.5882		27	485	FedAvg (mean over configurati	0.5893	-0.0011	yes

Fedavg Vs Fedprox Paired (columns 1/3)

split	hospitals	fedavg	fedprox	fedavg accuracy	fedprox accuracy	delta accuracy	fedavg macro precision	fedprox macro precision
als, balanced (50/50) -> 50.0%	2	test02	test03	0.4030	0.4328	0.0298	0.3915	0.4028
balanced (33.3 each) -> 33.3%	3	test04	test05	0.4851	0.4590	-0.0261	0.4333	0.4208
sizes matched to 3_clients_col	3	test12	test13	0.4478	0.4664	0.0186	0.4187	0.3925
each (DUKE I-SPY1 I-SPY2)	3	test10	test11	0.4291	0.4590	0.0299	0.3523	0.4302
ced (25 each) -> 25.0% / 25.0%	4	test06	test07	0.4776	0.4739	-0.0037	0.4646	0.4389
dissertation: 50/20/10/10) ->	4	test08	test09	0.4888	0.4515	-0.0373	0.4374	0.4365

Fedavg Vs Fedprox Paired (columns 2/3)

fedavg macro recall	fedprox macro recall	fedavg macro f1	fedprox macro f1	delta macro f1	fedavg macro auc	fedprox macro auc	Δ macro AUC	fedavg best round
0.3742	0.4025	0.3744	0.4001	0.0257	0.5594	0.5917	0.0323	25
0.4198	0.4127	0.4231	0.4116	-0.0115	0.5990	0.5958	-0.0032	27
0.4183	0.3885	0.4153	0.3884	-0.0269	0.5836	0.5882	0.0046	17
0.3582	0.4105	0.3536	0.4144	0.0608	0.5426	0.5678	0.0252	7
0.4522	0.4393	0.4378	0.4362	-0.0016	0.6531	0.6075	-0.0456	0
0.4259	0.4210	0.4292	0.4197	-0.0095	0.5982	0.6250	0.0268	21

Fedavg Vs Fedprox Paired (columns 3/3)

fedprox best round	fedavg training time s	fedprox training time s	within noise floor	favours
29	313	351	yes	—
28	313	333	yes	—
27	471	485	yes	—
16	469	482	yes	—
0	317	335	yes	—
2	323	387	yes	—

2 Hospitals (columns 1/2)

experiment	run	algorithm	hospitals	partition	split	accuracy	macro precision	macro recall
test01	test01_centralized	centralized	1	-	all data pooled	0.5299	0.4606	0.4503
test02	test02_fedavg_2h	fedavg	2	2_clients_balanced	2 hospitals, balanced (50/50) -> 50.0%	0.4030	0.3915	0.3742
test03	test03_fedprox_2h	fedprox	2	2_clients_balanced	2 hospitals, balanced (50/50) -> 50.0%	0.4328	0.4028	0.4025

2 Hospitals (columns 2/2)

macro F1	macro AUC	best epoch	best round	train time (s)	reference	reference macro AUC	Δ macro AUC	within noise floor
0.4523	0.6068	4		268	centralized (test01)	0.6068	0.0000	yes
0.3744	0.5594		25	313	centralized (test01)	0.6068	-0.0474	yes
0.4001	0.5917		29	351	centralized (test01)	0.6068	-0.0151	yes

3 Hospitals (columns 1/2)

experiment	run	algorithm	hospitals	partition	split	accuracy	macro precision	macro recall
test01	test01_centralized	centralized	1	-	all data pooled	0.5299	0.4606	0.4503
test04	test04_fedavg_3h	fedavg	3	3_clients_3hospitals	balanced (33.3 each) -> 33.3%	0.4851	0.4333	0.4198
test05	test05_fedprox_3h	fedprox	3	3_clients_3hospitals	balanced (33.3 each) -> 33.3%	0.4590	0.4208	0.4127
test10	test10_fedavg_cohort	fedavg	3	3_hospitals_cohort	one cohort each (DUKE I-SPY1 I-SPY2)	0.4291	0.3523	0.3582
test12	test12_fedavg_sizematched	fedavg	3	3_hospitals_sizematched	sizes matched to 3_clients_cohort	0.4478	0.4187	0.4183
test11	test11_fedprox_cohort	fedprox	3	3_hospitals_cohort	one cohort each (DUKE I-SPY1 I-SPY2)	0.4590	0.4302	0.4105
test13	test13_fedprox_sizematched	fedprox	3	3_hospitals_sizematched	sizes matched to 3_clients_cohort	0.4664	0.3925	0.3885

3 Hospitals (columns 2/2)

macro F1	macro AUC	best epoch	best round	train time (s)	reference	reference macro AUC	Δ macro AUC	within noise floor
0.4523	0.6068	4		268	centralized (test01)	0.6068	0.0000	yes
0.4231	0.5990		27	313	centralized (test01)	0.6068	-0.0078	yes
0.4116	0.5958		28	333	centralized (test01)	0.6068	-0.0110	yes
0.3536	0.5426		7	469	centralized (test01)	0.6068	-0.0642	yes
0.4153	0.5836		17	471	centralized (test01)	0.6068	-0.0232	yes
0.4144	0.5678		16	482	centralized (test01)	0.6068	-0.0390	yes
0.3884	0.5882		27	485	centralized (test01)	0.6068	-0.0186	yes

4 Hospitals Balanced (columns 1/2)

experiment	run	algorithm	hospitals	partition	split	accuracy	macro precision	macro recall
test01	test01_centralized	centralized	1	-	all data pooled	0.5299	0.4606	0.4503
test06	test06_fedavg_4h	fedavg	4	4_clients, balanced	all data pooled	0.4776	0.4646	0.4522
test07	test07_fedprox_4h	fedprox	4	4_clients, balanced	all data pooled	0.4739	0.4389	0.4393

4 Hospitals Balanced (columns 2/2)

macro F1	macro AUC	best epoch	best round	train time (s)	reference	reference macro AUC	Δ macro AUC	within noise floor
0.4523	0.6068	4		268	centralized (test01)	0.6068	0.0000	yes
0.4378	0.6531		0	317	centralized (test01)	0.6068	0.0463	yes
0.4362	0.6075		0	335	centralized (test01)	0.6068	0.0007	yes

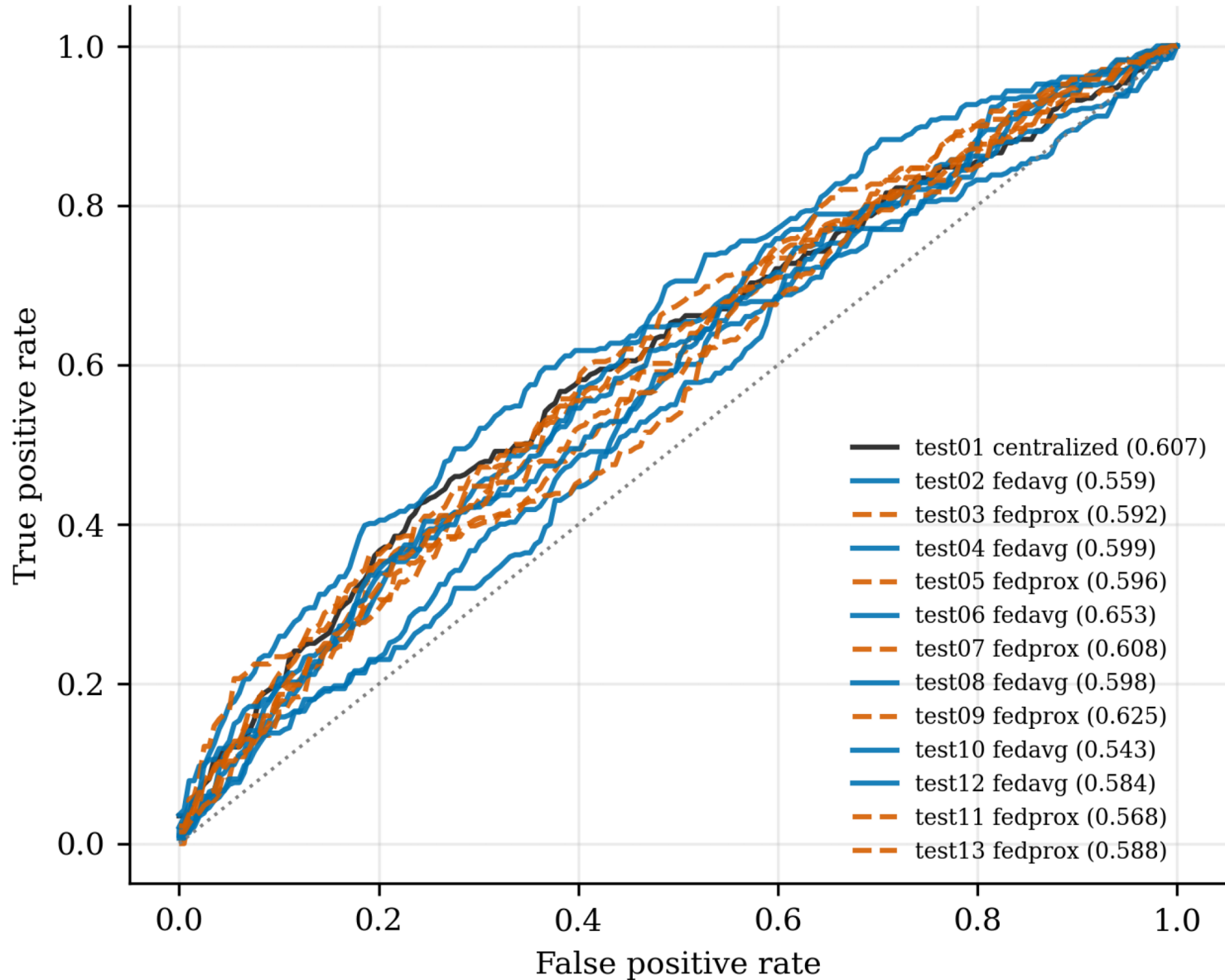
4 Hospitals Skewed (columns 1/2)

experiment	run	algorithm	hospitals		partition	split	accuracy	macro precision	macro recall
test01	test01_centralized	centralized	1		-	all data pooled	0.5299	0.4606	0.4503
test08	test08_fedavg_skewed	fedavg	4	4 hospitals skewed 50/50	4 hospitals skewed 50/50	all data pooled (dissertation: 50/20/10/10) ->	0.4888	0.4374	0.4259
test09	test09_fedprox_skewed	fedprox	4	4 hospitals skewed 50/50	4 hospitals skewed 50/50	all data pooled (dissertation: 50/20/10/10) ->	0.4515	0.4365	0.4210

4 Hospitals Skewed (columns 2/2)

macro F1	macro AUC	best epoch	best round	train time (s)	reference	reference macro AUC	Δ macro AUC	within noise floor
0.4523	0.6068	4		268	centralized (test01)	0.6068	0.0000	yes
0.4292	0.5982		21	323	centralized (test01)	0.6068	-0.0086	yes
0.4197	0.6250		2	387	centralized (test01)	0.6068	0.0182	yes

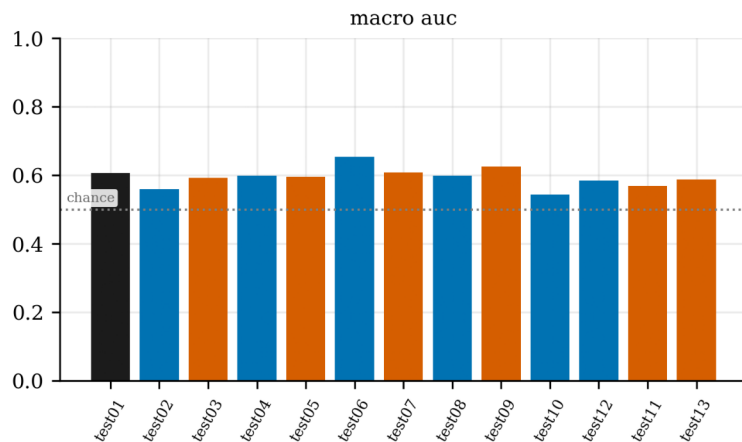
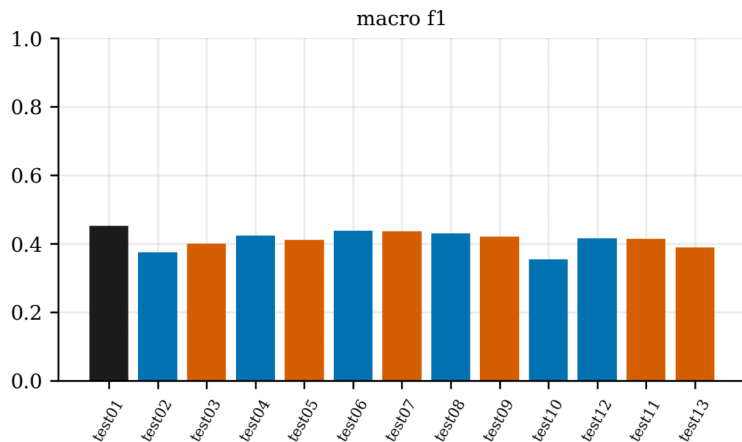
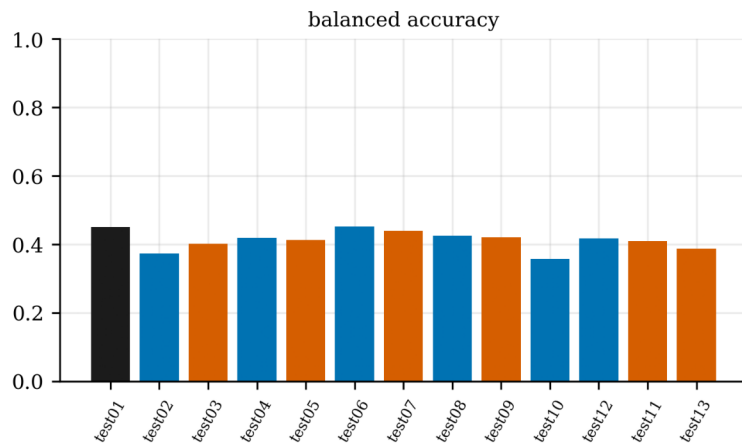
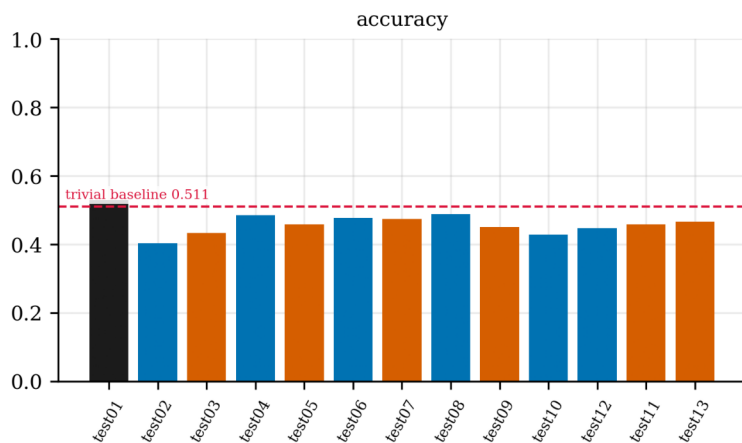
Macro-average ROC — all experiments, same global test set



bar metrics all

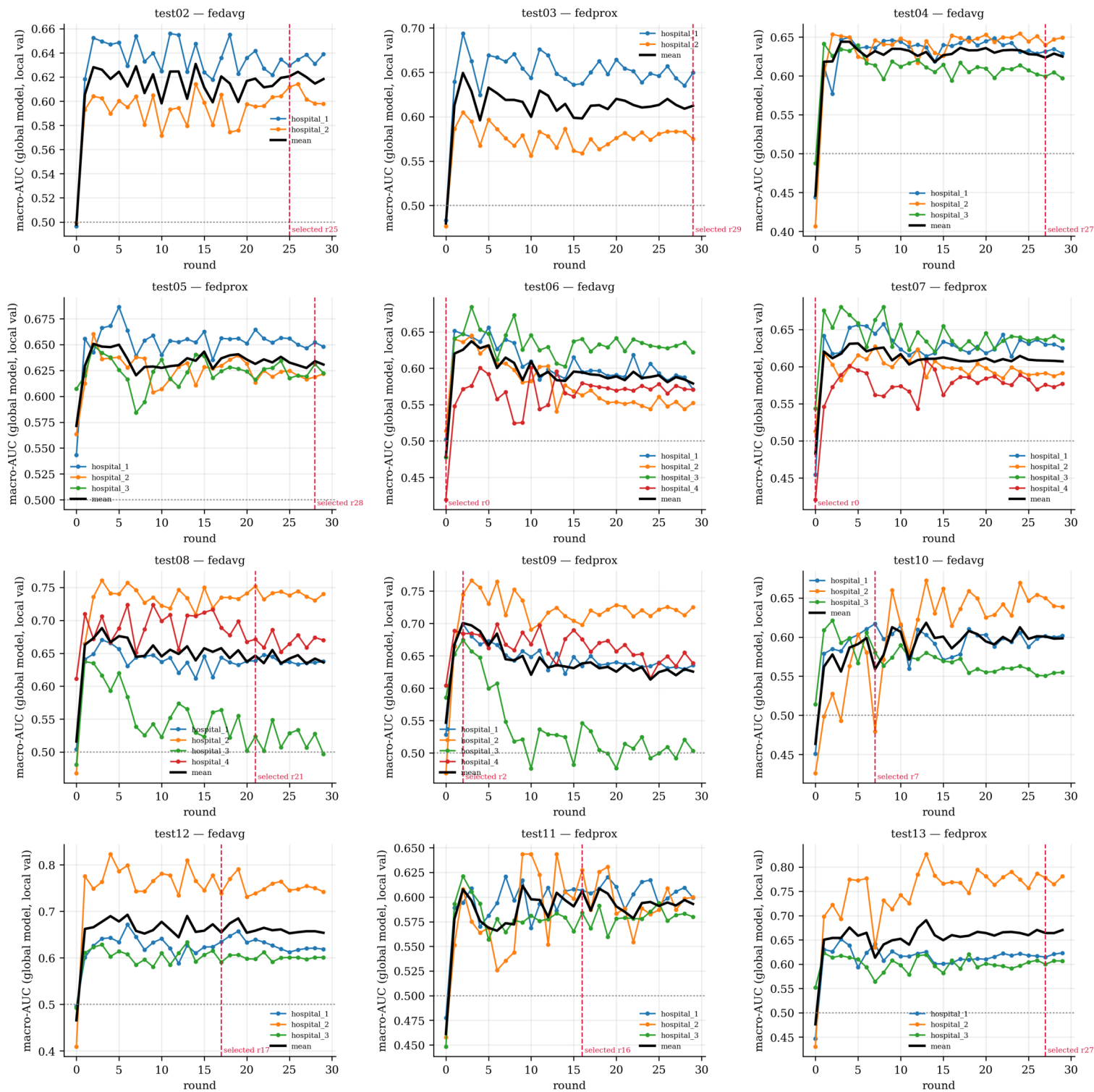
All experiments on the global test set

centralized fedavg fedprox

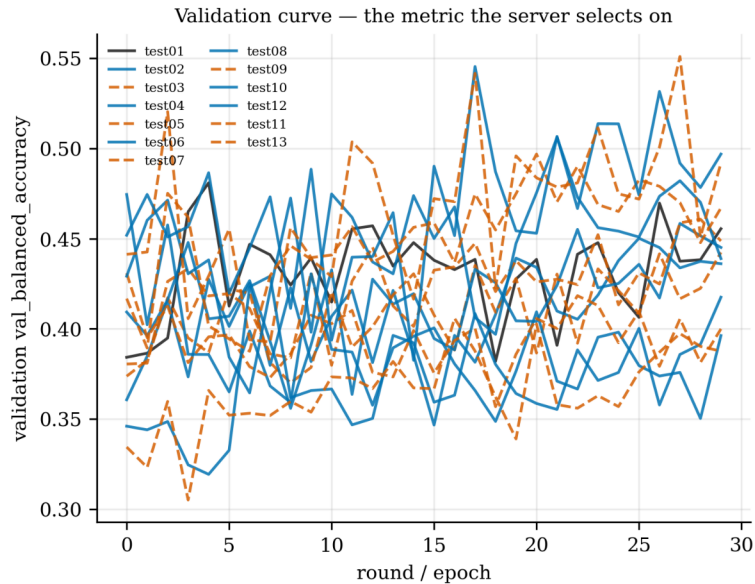
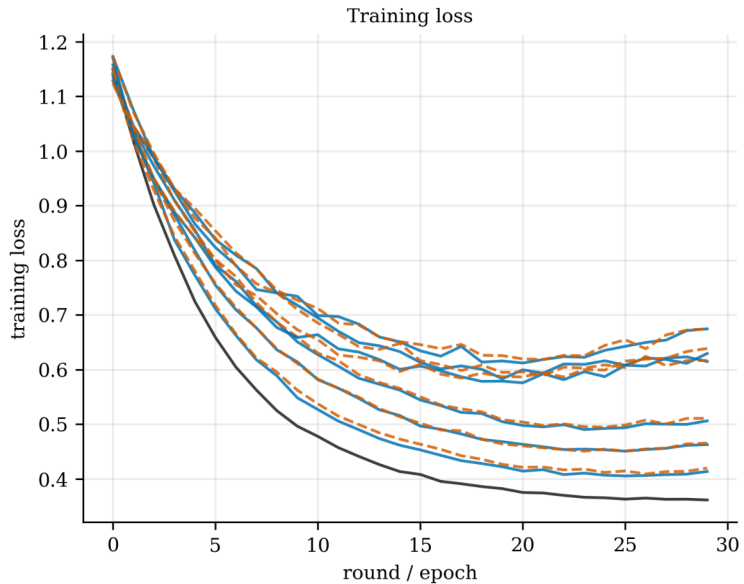


federated round evolution

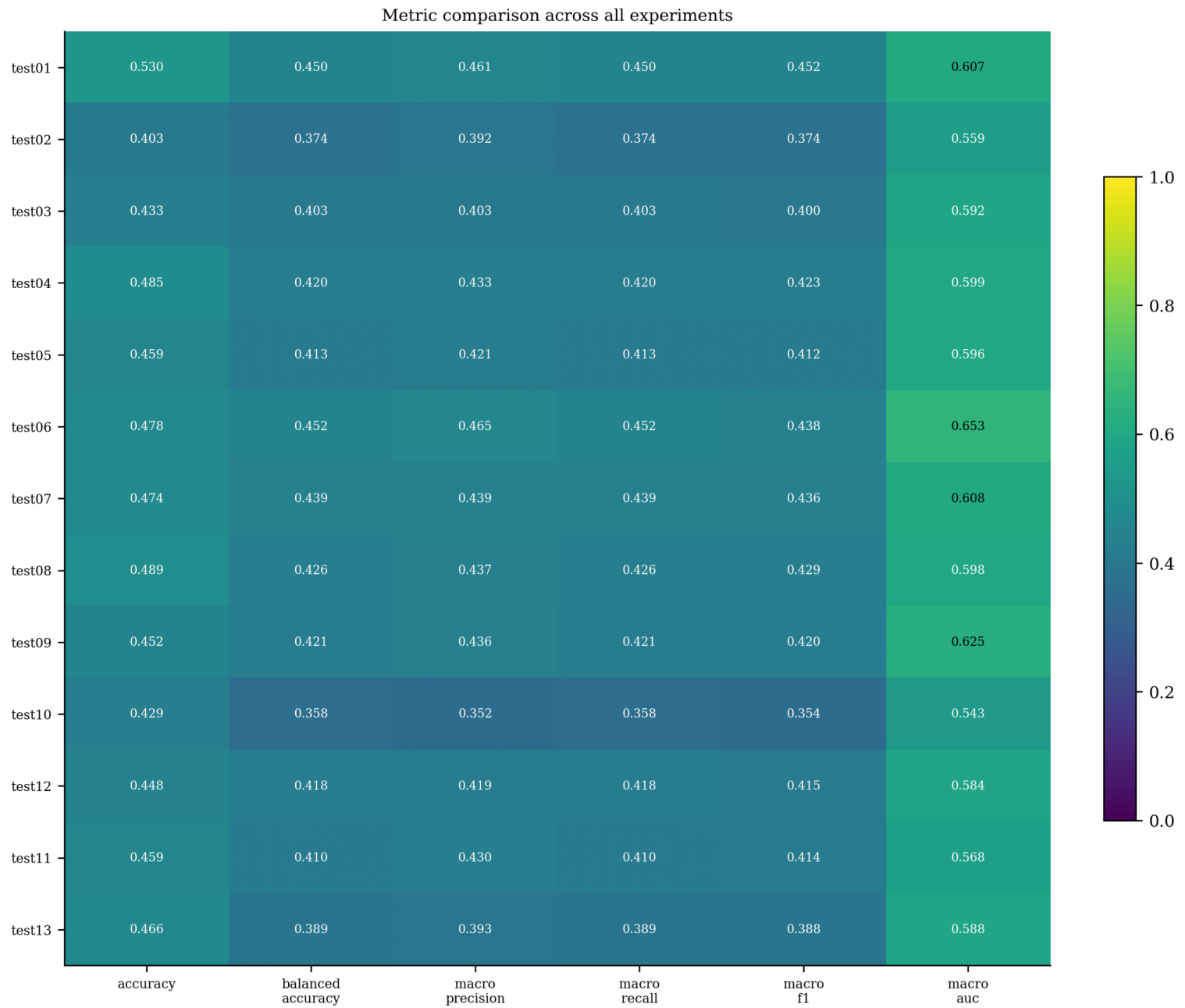
Federated round evolution — the arriving global model scored at each hospital



training evolution



metric comparison heatmap



class distribution

Class distribution per hospital — stratified, so only QUANTITY varies

